AdifySkills

n8n Automation Mastery
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MENTOR COPY

Module 2 · Core Nodes & Data Mastery

DAY 06 of 25

Code Node: JavaScript for Automators

"Jahan node khatam, wahan Code node shuru."

⚡ AAJ HUM YEH BANAYENGE

GST Invoice Calculator (Code Node)

Order ke items se subtotal, discount, GST (CGST+SGST) aur grand total JavaScript se calculate karke ek formatted invoice JSON file banata hai.

Code (JS) | Code (Python) | AI code helper

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Code node se student "no-code ki limit" tod deta hai. Thodi si JavaScript = unlimited power. AI se code likhwana bhi sikhaate hain.

🎯 Aaj ke baad aap yeh kar paoge

01 JavaScript essentials for n8n (no CS degree needed)

🌍 Recipe mein ingredients (variables), "agar namak kam ho to daalo" (if), "5 baar hilao" (loop).
⚙️ const total = $json.price * $json.qty; if (total > 1000) { ... }

02 Run once for all items vs each item

🌍 Poori class ki copies ek saath check vs har bachche ko alag bulana.
⚙️ All items: $input.all() · Each item: $json

03 Return format: [{json: {...}}]

🌍 Courier ka fixed packing format — galat packing = reject.
⚙️ return [{json: {total: 500}}];

04 Array methods: map, filter, reduce

🌍 map = har aam ko kaatna, filter = kharab aam hatana, reduce = sabka juice ek glass mein.
⚙️ items.map(i => i.price), items.filter(i => i.qty > 0), items.reduce((s, i) => s + i.price, 0)

05 Using AI to write & debug Code node

🌍 Tuition teacher se homework samajhna — copy nahi, samajh ke likhna.
⚙️ Code node mein "Ask AI": "GST 18% add karo aur 2 decimal mein round karo".

06 Python in Code node — basics

🌍 Hindi aur English — dono se baat ho sakti hai.
⚙️ Python mode: for item in _input.all(): item.json["x"] = 1

⏱ Class ka plan (2 ghante)

10m	40m	40m	20m	10m

0:00–0:10 Recap

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

👩‍🏫 2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

0:10–0:50 Concept: JS Crash Course

Variables, objects, loops — n8n context

🪄 Recipe likhna jaisa hai: pehle ingredients lo, phir step-by-step karo, end mein plate mein serve karo — yahi return hai.

👩‍🏫 Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

0:50–1:30 Live Build

Invoice line-item calculator + GST

🪄 Aaj ki kahani yaad hai — 'Dukaan ka hisaab — calculator wala munshi'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

👩‍🏫 Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

1:30-1:50 **Quiz + Challenge**

Module 2 quiz

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

👩‍🏫 Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

1:50-2:00 **Q&A + Homework**

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

👩‍🏫 Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Run once for all items

Saare items ek saath milte hain: $input.all()

🪄 **Analogy:** Poori class ki attendance ek baar mein

CONCEPT 02

Run once for each item

Har item par alag chalta hai: $json

🪄 **Analogy:** Har student se alag baat

CONCEPT 03

Return format

Hamesha array of {json: {...}} return karo.

🪄 **Analogy:** Packing format fixed hai

CONCEPT 04

map / filter / reduce

Arrays ke 3 superpowers — transform, chhaanto, jodo.

🪄 **Analogy:** Factory machines

🔑 Aaj ka main idea

Code node mein thoda JavaScript likh ke woh kaam karo jo 5–6 nodes mein hota. Bas 3 cheezein yaad rakho: data $input/$json se lo, map/filter/reduce se badlo, aur [{json: {...}}] return karo. AI se code likhwana bhi allowed hai — samajhna zaroori hai.

- map: har item ka price double karna
- filter: sirf 18+ age wale
- reduce: saare amounts ka total

📖 Ek kahani se samjho: Dukaan ka hisaab — calculator wala munshi

1 🧾 **Order list aayi** = Webhook

2 👨‍💻 **Munshi ne hisaab lagaya** = Code node

3 ➕ **Total + GST + discount** = reduce() / math

4 🏷️ **Bill number diya** = Edit Fields

5 ↩️ **Customer ko bill** = Respond to Webhook

🪄 Ek line mein: “Recipe likhna jaisa hai: pehle ingredients lo, phir step-by-step karo, end mein plate mein serve karo — yahi return hai.”

Aaj ka workflow: GST Invoice Calculator (Code Node)

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Shops, small manufacturers (GST invoice calculator) — aam taur par **₹5,000 – ₹15,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ Webhook
Order aata hai with line items

2 👨‍💻 Code
Har item ka subtotal, discount, 18% GST, grand total

3 ✏️ Edit Fields (Set)
invoiceNo = "INV-" + $now.toFormat("yyMMdd") + random

4 ⚡ Respond to Webhook
Calculated invoice JSON return

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- ☐ Koi setup nahi — seedha Test karo!

✅ Test kaise karein

1. "Test workflow" dabao
2. "Calculate Invoice" output mein har line ka GST check karo
3. Last node mein "Download" se invoice.json lo

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Calculator se har item jodna	10 min
GST alag nikaalna	5 min
Galti dobara check	5 min
	20 min

⚡ Ab (n8n)

Code node 1 second mein

0 min

⏱ Har bill par 20 min bache, calculation galti zero.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo 📋 Select karke copy karo

```
const o = $input.first().json;
const subtotal = o.items.reduce((s, i) => s + i.price * i.qty, 0);
const discount = subtotal * (o.discount / 100);
const gst = (subtotal - discount) * 0.18;
return [{ json: { subtotal, discount, gst: +gst.toFixed(2), total: +(subtotal - discount +
gst).toFixed(2) } }];
```

📦 Sample data — input se output tak

📥 Input 📋 Select karke copy karo

```
{
  "items": [
    {
      "name": "Course",
      "price": 9999,
      "qty": 1
    },
    {
      "name": "Templates",
      "price": 1999,
      "qty": 2
    }
  ],
  "discount": 10
}
```

📤 Output 📋 Select karke copy karo

```
{
  "subtotal": 13997,
  "discount": 1399.7,
  "gst": 2267.51,
  "total": 14864.81
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Order ke items se subtotal, discount, GST (CGST+SGST) aur grand total JavaScript se calculate karke ek formatted invoice JSON file banata hai.

Flow: Manual Trigger (Test run) → Code (Sample order) → Code (GST + discount + total) → Convert to File (invoice.json)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "GST Invoice Calculator (Code Node)" (nodes: Run Manually, Sample Order,
Calculate Invoice, Make invoice.json).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
4 nodes · ready to import
`adifyskills.com/workflows/day-06-gst-invoice-calculator.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/6`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo

⚠️ Yeh galtiyan sabse common hain

- ! Object return karna array ki jagah
- ! "each item" mode mein $input.all() use karna
- ! Floating point rounding bhool jaana (toFixed)

⭐ Pro tips — time bachane wali tricks

- ★ Code node mein "Ask AI" se code generate karwao, phir samjho
- ★ console.log() output browser console mein dikhta hai
- ★ Complex logic ke liye pehle 5–6 nodes se banao, phir Code mein compress

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Order items se GST, discount aur total calculate karke formatted invoice JSON banana.

📝 Homework

3 tasks jo pehle 4–5 nodes mein hote the, unhe ek Code node mein likho.

- [ ] Homework ho gaya

BEGINNER

Naam ko uppercase karo Code se

- [ ]

INTERMEDIATE

Duplicate emails Code se hatao

- [ ]

ADVANCED

Indian number format (1,00,000) function likho

- [ ]

🧩 Aaj ka challenge — pehle khud try karo

🎯 Agar subtotal ₹2,000 se upar ho to free shipping, warna ₹99 shipping jodo.

💡 Hint 1

Grand total ke calculation mein ek naya variable banao.

💡 Hint 2

const shipping = subtotal > 2000 ? 0 : 99; — yeh ternary operator hai.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Code node ka sahi return?

(A) "a"
(B) true
(C) {a:1}
(D) [{json:{a:1}}]

Q2. Saare items kaise milte hain?

(A) $input.all()
(B) $items
(C) $all
(D) $json

Q3. Sum nikaalne ke liye array method?

(A) map
(B) filter
(C) reduce
(D) sort

Q4. Code node kin languages mein?

(A) PHP
(B) Sirf JS
(C) JS + Python
(D) Sirf Python

📖 Aaj ke naye shabd

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana…).

Webhook
Ek URL jis par koi doosra app data bhejta hai aur aapka workflow turant chal padta hai.

JSON
Data likhne ka format: { "naam": "Ravi", "city": "Delhi" } — key aur value ke jode.

Item
Data ki ek row/record. n8n mein data items ki list ki tarah flow karta hai.

Code Node
Jahan aap JavaScript likh ke apna custom logic chala sakte ho.

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👨‍🏫 Mentor notes — class kaise lein

Yeh page sirf aapki copy mein hai. Students ki PDF mein nahi jaata.

- [ ] "Code se mat daro" — pehle line-by-line Hinglish mein padhao
- [ ] AI se code likhwa ke usme jaan-boojh kar bug daalo, students dhoondhein
- [ ] Module 2 quiz

🗣️ Har block ke liye mentor tip

10m — Recap

2 students se homework screen share karwao. Galti wala homework bhi dikhao — sabse zyada seekh wahin se aati hai.

Class se poochho: Kal wale workflow mein sabse mushkil step kaunsa laga? · Homework mein kahan error aaya aur kaise theek kiya? · Kal ka ek node naam batao aur woh kya karta hai?

40m — Concept: JS Crash Course

Pehle real-life example se shuru karo, phir n8n screen dikhao. Har concept ke baad 1 student se apne shabdon mein dobara bulwao.

Class se poochho: Code node ka return format kya hota hai? · Saare items kaise milte hain? · Total nikaalne ka method?

40m — Live Build

Har 10 minute par poocho "sabka green tick aaya?". Jaan-boojh ke ek galti live hone do aur use debug karke dikhao.

Class se poochho: Is node ka output agle node mein kaise ja raha hai? · Agar yeh node fail ho to kya hoga? · Isko apne kaam/business mein kahan use kar sakte ho?

20m — Quiz + Challenge

Pehle 5 minute akele try karne do, phir hint do. Jo solve kare use screen share karwao.

Class se poochho: Code node ka sahi return? · Saare items kaise milte hain? · Sum nikaalne ke liye array method?

10m — Q&A + Homework

Chat se top 3 doubts live solve karo. Homework ka pehla step class mein hi karwa do, taaki koi blank na baithe.

Class se poochho: Aaj ka sabse useful node kaunsa laga? · Homework mein pehla step kya hoga? · Aaj ki class 1 line mein summarize karo.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 7

HTTP Request & REST APIs Mastery

"Agar API hai, toh n8n se connect ho sakta hai."

☐ **Day 6 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

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🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → D
[{ json: { a: 1 } }]

Q2 → A
$input.all()

Q3 → C
reduce

Q4 → C
JS + Python

✅ Challenge ka solution

Calculate Invoice mein: const shipping = sum('base') > 2000 ? 0 : 99; aur grand_total mein + shipping, output mein shipping field bhi do.

📚 Official docs

- Code node ↗
- Cookbook ↗
- MDN JS Guide ↗
- Cookbook: Code node ↗
- AI coding in Code node ↗
- MDN: Array.reduce ↗

▶ Videos

- n8n code node javascript tutorial ↗
- n8n code node python ↗

Doubt poochho
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